



THE 2026 GUIDE TO SOLVING CONNECTIVITY CHALLENGES IN ELECTRONIC WARFARE.



Elsight Halo for Military Comms Personnel

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In modern military and tactical operations, persistent, secure, and intelligent connectivity is mission-critical. From SOF and intelligence missions to autonomous reconnaissance, uninterrupted real-time situational awareness and robust command-and-control (C2) connections are essential.

Elsight Halo with its multi-network link bonding ensures robust, high-bandwidth connectivity even in the most contested environments.

Originally designed for BVLOS drone commercial operations, Halo has evolved into a tactical-grade solution bridging NLOS (non-line-of-sight) communication gaps for all unmanned systems.

Soldiers operating in an electronic warfare (EW) environment face a range of complex and evolving challenges. These challenges affect all domains of warfare from communications, navigation, command and control, and in particular affects situational awareness. Here are some examples of major communications challenges that soldiers can encounter:

Communication Disruptions

One of the most immediate effects of EW is the jamming or interception of communications preventing units from coordinating movements or calling for support. Others include:

- Spoofing attacks can inject false messages or commands that can mislead forces into unsafe or strategically unsound positions.
- Reliance solely on line-of-sight radios becomes risky when frequencies are under attack:

Frequently, a target of EW is the Intelligence, Surveillance, and Reconnaissance (ISR) platforms like drones, radars, and satellites to eliminate data transmission from situational awareness missions.

All these challenges and others cause mounting problems for deployed forces and unmanned systems, translating to loss of Command and Control (C2) connections, interrupted ISR feeds and the inability to reroute or repurpose assets in real-time.

Adding a Critical Link to Traditional Communications

Elsight's Halo palm-sized, 93 grams, connectivity platform ensures mission continuity by dynamically bonding multiple comms links using LTE or 5G, RF and satellite communications, into one bonded, secure channel. Halo continuously monitors link performance and prioritizes its steering routes according to the lowest latency and highest reliability.

Deployment Versatility and Edge-Ready Integration

Halo's low SWaP (Size, Weight, and Power) design is a critical enabler for tactical flexibility. It can be integrated into unmanned systems of many formats, including:

- Unmanned Aerial Systems (UAVs) such as those used for ISR missions and logistics
- Unmanned Ground Vehicles (UGV) such as those used for logistics or perimeter patrol in denied terrain

In contested environment conditions, mobility and modularity are key. Halo is platform-agnostic and NDAA-compliant, battlefield proven in constrained spaces, difficult terrain and extreme situations.



Operational Demonstrations: Resilience Under Adversity

Scenario A: An allied SOF unit executed a simulated reconnaissance mission in a RF-contested valley

Mission Context:

In a recent joint special operations field validation, an allied SOF unit executed a simulated reconnaissance mission in a RF-contested valley modelled after a known high-threat area. The objective was to demonstrate continuous intelligence gathering and command-and-control (C2) in an environment replicating the adversary's advanced electronic warfare (EW) capabilities.

Operational Challenge: The mission required persistent aerial ISR from a small, unmanned aircraft system (sUAS) providing HD electro-optical/infrared (EO/IR) video feeds to both forward-deployed vehicles and a remote command post. The area's steep terrain and deliberate RF jamming created conditions in which traditional single-link communications solutions would have failed, risking mission success and operator safety.

Elsight Halo Deployment: To ensure resilience, a vehicle-mounted Halo node served as both a mobile C2 hub and a mesh expansion point, connecting airborne ISR assets, forward elements on foot, and higher headquarters. The system was configured to aggregate SATCOM and LTE connectivity from multiple carriers, automatically optimizing link usage in real time.

The operational stressors included targeted RF jamming, degrading both primary LTE bands and satcom intermittently.

Despite these challenges, the Halo network maintained uninterrupted bi-directional data flow between the aerial ISR platform, the vehicle node, and the command center, enabling live intelligence exploitation, rapid decision-making, and simulated mission success.

Scenario B: Special Forces Reconnaissance in a Communications-Denied Environment

Mission Context:

A joint NATO special forces team is deployed to a remote, mountainous border region to conduct covert reconnaissance on suspected insurgent weapons depots. The area is known for aggressive EW activity such as the enemy forces deploying mobile jammer and cyber intrusion tools to disrupt command-and-control (C2) communications. Traditional satcom links are unreliable due to topographical masking and heavy cloud cover.

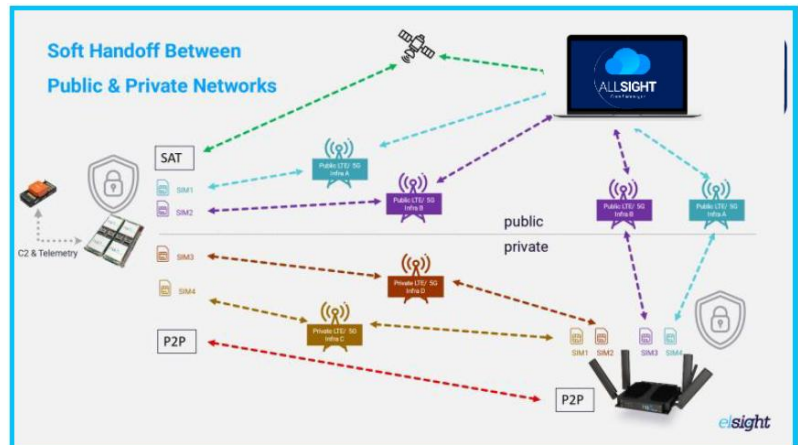
Operational Challenge: The team needs persistent, real-time connectivity to coordinate with headquarters, stream ISR drone feeds, and update tactical maps. Any communications blackout would risk mission failure and jeopardize operator safety.

Elsight Halo Deployment:

- **Configuration:** Each operator's kit included a Halo unit integrated into their tactical radio, while unmanned ground vehicles (UGVs) and rotary-wing drones are also Halo-enabled.
- **Connectivity strategy:** Halo aggregated and encrypted multiple links simultaneously, leveraging cellular (across different national carriers), line-of-sight radio relays, and LEO satellite modems to form a bonded, self-healing, low-latency connection.
- **Adaptive Resilience:** As jammers intermittently targeted one or more bands, Halo's intelligent link-management rerouted data packets through surviving channels in milliseconds, with no perceptible drop in video or voice quality.

Scenario B Results: Real-time ISR streaming from drones to the command post continues uninterrupted, enabling analysts to identify the exact location of two hidden weapons caches.

- Continuous tactical coordination allowed the team to alter their approach route in response to enemy patrol movements detected on live video.
- UGVs provided live forward reconnaissance without risk to personnel, thanks to the Halo's bonded links maintaining control signals despite EW attacks.
- The mission was completed without compromise, target coordinates were relayed instantly to allied air assets, leading to precision strikes within minutes.



Why the Halo Was Critical: In this denied environment, no single comms channel would survive consistently, but Halo's multi-link aggregation and high redundancy preserves operational tempo and situational awareness. Without Halo, the unit would have been blind and forced to abort.

Conclusion: Elsight Halo as a Tactical Enabler

Elsight's Halo is a disruptor among the traditional radio or mesh communications with embedded multilink cellular networks, enabling resilient communications for UAVs and UGVs. The Halo is integrated and battlefield proven in conflicts around the world enabling uninterrupted C2 and real time ISR data transfer even in war zones. Halo's primary advantage is its smart multilink bonding with embedded cellular links, with dynamic data traffic management to provide real-time C2, telemetry and video transmissions without relying on bulky high-power radios.

Elsight Halo transforms the concept of connectivity from a vulnerability to a force multiplier. In missions where failure is not an option, Halo ensures secure, persistent connectivity, failover-proof live video and data streams, and operational agility in comms-contested environments. Whether extending the eyes of a command center or enabling drones to fly where humans can't, Halo delivers. Halo ensures your communications are as resilient as your mission demands.

Product Feature	Value
Comms Channels	LTE, 5G, SATCOM, RF
Latency	Sub-200ms failover
Integration	Platform-agnostic
Compliance	NDAA, MIL-STD options available
Security	AES-256 end-to-end encryption
Dimensions	Compact SWaP-C optimized – 93 grams, 86 x 69 x14 mm

About Elsight

Elsight delivers Connection Confidence for Unmanned Aerial and Unmanned Ground Vehicles (UAVs, UGVs) collectively logging more than 400,000 of operational hours. The company's customers span the globe and include UAV/UGV manufacturers, operators, and integrators for commercial, defense and HLS/public safety markets.